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0.4 | Equivalent Viscous Damping

Equivalent viscous damping is the simplest form of damping for analysis since the governing differential equation of motion is linear and hence amenable to analytical solution. This section determines the damping coefficient for viscous damping so that it is equivalent in some sense to the combined effect of all damping mechanisms present in the actual structure. [0.4.1]

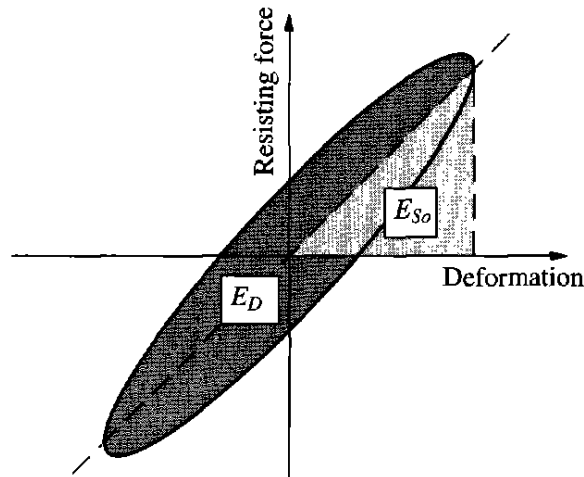


Fig. 1 - Ideal Viscous Damping Model

The simplest definition of equivalent viscous damping is based on the measured response of a system to harmonic force at exciting frequency ω equal to the natural frequency ω_n of the system. The damping ratio ζ_{eq} is calculated from using measured values of u_o and $(u_{st})_o$. This is the equivalent viscous damping since it accounts for all the energy-dissipating mechanisms that existed in the experiments.

0.4 - 2 | Measured Viscous Damping

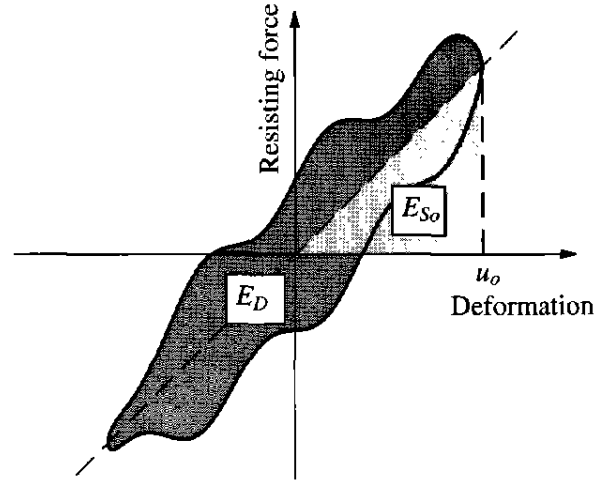


Fig. 2 - Measured Viscous Damping Model

The most common method for defining equivalent viscous damping is to equate the energy dissipated in a vibration cycle of the actual structure and an equivalent viscous system. For an actual structure the force-displacement relation obtained from an experiment under cyclic loading with displacement amplitude u_o is determined; such a relation of arbitrary shape is shown schematically in Fig. 2. The energy dissipated in the actual structure is given by the area E_D enclosed by the hysteresis loop. Equating this to the energy dissipated in viscous damping given by Eq.1 leads to

Eq.1

$$4\pi\zeta_{eq}\frac{\omega}{\omega_n}E_{So} = E_D$$

or, equivalently,

Eq.2:

$$\zeta_{eq} = 1 \cdot \frac{1}{4 \cdot \pi} \cdot 1 \cdot \frac{1}{\left(\frac{\omega}{\omega_n}\right)} \cdot E_D \cdot \frac{1}{E_{So}}$$

where the strain energy, $E_{So} = ku_o^2/2$, is calculated from the stiffness k determined by experiment.

The experiment leading to the force-deformation curve of Fig. 3.9.1 and hence E_D should be conducted at $\omega = \omega_n$, where the response of the system is most sensitive to damping. Thus Eq. specializes to

Eq.3:

$$\zeta_{eq} = 1 \cdot \frac{1}{4 \cdot \pi} \cdot E_D \cdot \frac{1}{E_{So}}$$

The damping ratio ζ_{eq} determined from a test at $\omega = \omega_n$ would not be correct at any other exciting frequency, but it would be a satisfactory approximation (Section 3.10.2).

It is widely accepted that this procedure can be extended to model the damping in systems with many degrees of freedom. An equivalent viscous damping ratio is assigned to each natural vibration mode of the system (defined in Chapter 10) in such a way that the energy dissipated in viscous damping matches the actual energy dissipated in the system when the system vibrates in that mode at its natural frequency.

In this book the concept of equivalent viscous damping is restricted to systems vibrating at amplitudes within the linearly elastic limit of the overall structure. The energy dissipated in inelastic deformations of the structure have also been modeled as equivalent viscous damping in some research studies. This idealization is generally not satisfactory, however, for the large inelastic deformations of structures expected during strong earthquakes. We shall account for these inelastic deformations and the associated energy dissipation by nonlinear force-deformation relations, such as those shown in Fig. 1.3.4 (see Chapters 5 and 7).

[0.4.1]

Anil K. Chopra, Dynamics of Structures: Theory and Applications to Earthquake Engineering. Englewood Cliffs, NJ, USA: Prentice Hall, 1995.

